107. Which of the following pairs is/are correctly matched?

Inventor		Invention
1. Christopher Cockerell	:	Hovercraft
2. David Bushnell	:	Submarine
3. J.C. Perrier	:	Steamship

Select the correct answer by using the code given below :

(a) 1, 2 and 3 (b) 1 and 2, only
(c) 2 and 3, only (d) 1 and 3, only

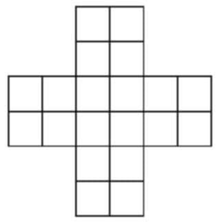
Ans. (a) : The Hovercraft invented by Christopher Cockerell is officially launched in Southampton. Sir Christopher Cockerell invented the hovercraft, the vehicle which can travel across land or water on a cushion of air. David Bushnell is credited as the inventor of the Submarine, which was first used to launch explosives against British ships during the American Revolution (1775-83). Steam ship was invented by J.C Perrier in the year 1775.

108. Who among the following wrote "The Communist Manifesto" along with Karl Marx?

(a) Emile Durkheim
(b) Friedrich Engels
(c) Robert Owen
(d) Max Weber

Ans. (b) : The Communist Manifesto was written by Karl Marx with the assistance of Friedrich Engels.

109.



Each of 8 identical balls is to be placed in the squares shown in the figure given above in a horizontal direction such that one horizontal row contains 6 balls and the other horizontal row contains 2 balls. In how many maximum different ways can this be done?

(a) 38 (b) 28
(c) 16 (d) 14

Ans. (a) : There can be two cases:
(i) when x row contains 6 balls:
Then the 2 balls can be arranged in y row in $6p_2$ ways = 15 or the 2 balls can be arranged in any of he 4 two box row in 4 ways.
So, no of ways, when x contains 6 balls = 15 + 4 = 19
(ii) Similarly, no. of ways, when y row contains 6 balls = 19
As, either of case (i) or case (ii) is possible,
Hence, total no. of ways = 19 + 19 = 38 Ans.

110. Who is Wole Soyinka?

(a) A well-known economist
(b) A well-known football player
(c) A well-known industrialist owning steel plants in many parts of the world
(d) A Nobel Prize winner for literature

Ans. (d) : Akinwande Oluwole "Wole" Babatunde Soyinka is a Nigerian playwright and poet. He was awarded the 1986 Nobel Prize in Literature, the first African to be honored in that category.

111. In a tournament each of the participants was to play one match against each of the other participants. 3 players fell ill after each of them had played three matches and had to leave the tournament. What was the total number of participants at the beginning, if the total number of matches played was 75?

(a) 8 (b) 10
(c) 12 (d) 15

Ans. (d) : Let the total no. of participants be x at the beginning.
After 3 players fell in, remaining participants = $x - 3$
Now,

$$^{x-3}C_2 + 3 \times 3 = 75$$

$$\frac{(x-3)!}{2!(x-5)!} + 9 = 75$$

$$x^2 - 7x - 120 = 0$$

$$(x + 8)(x - 15) = 0$$

Neglecting the negative value of x we get
$x = 15$

112. P, Q, R, S and T reside in a 5-storeyed (Ground + 4) building and each of them resides on a separate floor. Further,
1. T does not reside on the topmost floor.
2. Q does not reside on the ground floor.
3. S resides on one storey above that of P and one storey below that of R.
To know as to which one of the 5 persons resides on the ground floor which of the above statements are sufficient/insufficient?

(a) 1 and 3 are sufficient
(b) 2 and 3 are sufficient
(c) 1, 2 and 3 are sufficient
(d) 1, 2 and 3 are insufficient

Ans. (d) : On the basis of the facts given in the question, there can be 3 arrangements.

①	②	③
R	Q	Q
S	R	T
P	S	R
Q	P	S
T	T	P

As either of T or P can reside on the ground floor, so to exactly find who resides on the ground floor, given statements are insufficient.

113. A box contains 5 sets of balls while there are 3 balls in each set. Each set of balls has one colour which is different from every other set. What is the least number of balls that must be removed from the box in order to claim with certainty that a pair of balls of the same colour has been removed?

(a) 6 (b) 7
(c) 8 (d) 9

Ans. (a) :

Set	1	2	3	4	5
Balls	1	1	1	1	1

Now any further removal of balls from any set will ensure that removed ball is of the same colour as one of the already removed balls, thus constituting a pair of the removed balls of the same colour.
Hence, minimum no. of removed balls = 6

114. There are three parallel straight lines. Two points A and B are marked on the first line, points C and D are marked on the second line and points E and F are marked on the third line. Each of these 6 points can move to any position on its respective straight line.
Consider the following statements:
1. The maximum number of triangles that can be drawn by joining these points is 18.
2. The minimum number of triangles that can be drawn by joining these points is zero.
Which of the statements given above is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (b) : Maximum number or triangles can be formed by selecting 3, 4 or 5 points out of 6 at a time.
$\therefore$ Maximum number of triangles = $^6C_3 + {}^6C_4 + {}^6C_5 = 41$
Which is more than 18. Hence 1st statement is incorrect.
Now, minimum no. of triangle can be zero, when the points overlap on the same line and also when all points are along the same vertical line. Hence 2nd statement is correct.

115. A mixed doubles tennis game is to be played between two teams (each team consists of one male and one female.) There are 4 married couples. No team is to consist of a husband and his wife. What is the maximum number of games that can be played?

(a) 12 (b) 21
(c) 36 (d) 42

Ans. (d) : Let married couples are –

	MF	MF	MF		MF
	ab	cd	ef	and	gh
$\therefore$ possible teams	ad	cb	eb	gb	af	cf	ed	gd
	ah	ch	eh	and	gf			

Now team ad can play with cd, cg, cb, eb, eh, gb and gf i.e. 7
The same will be the case with other teams i.e. 7
So no of total matches $12 \times 7 = 84$
Since every match includes two teams,
The number of matches = $\frac{84}{2} = 42$

116. In an office, the number of persons who take tea is twice the number of persons who take only coffee. The number of persons who take coffee is twice the number of persons who take only tea.
Consider the following statements:
1. The sum of the number of persons who take either tea or coffee or both is four times the number of persons who take both coffee and tea.
2. The sum of the number of persons who take only coffee and those who take only tea is twice the number of persons who take both tea and coffee.
Which of the statements given above is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

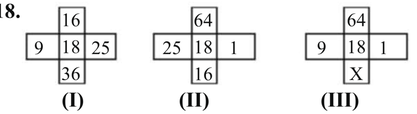
Ans. (c) : Let x number of persons take tea,
y number of persons take coffee and
z number of persons take both coffee and tea.
Now,
x + z = 2y .....(i)
y + z = 2x .....(ii)
From equation (i) and (ii)
x = y = z
So
$x + y + z = z + z + z = 3z$
Thus 1st statement is wrong
Again
$x + y = z + z = 2z$
Hence 2nd statement is correct.

117. 3 digits are chosen at random from 1, 2, 3, 4, 5, 6, 7, 8 and 9 without repeating any digit. What is the probability that their product is odd?

(a) 2/3 (b) 5/108
(c) 5/42 (d) 7/48

Ans. (c) : Total number of 3-digit no. = $9 \times 8 \times 7 = 504$
For product to be odd, we have to choose only from odd numbers.
$\therefore$ Total no. of 3-digit no. whose product are odd
$= 5 \times 4 \times 3 = 60$
$\therefore$ Required probability = $\frac{60}{504} = \frac{5}{42}$

118.



(I) (II) (III)

What is the value of X in figure III?